THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · EASY LEVEL UNIT TEST SET 3

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The value of $2x + 7$ when $x = 6$ is: 1

(a) 19 **(b)** 21 **(c)** 38 **(d)** 17

2. The area of a triangle with base 12 cm and height 7 cm is: 1

(a) 42 cm² **(b)** 84 cm² **(c)** 19 cm² **(d)** 54 cm²

3. The number of faces of a pentagonal prism is: 1

(a) 10 **(b)** 7 **(c)** 5 **(d)** 6

4. The mode of 14, 4, 7, 20, 8, 7, 7 is: 1

(a) 8 **(b)** 7 **(c)** 20 **(d)** 4

5. On simplifying $3x + 7 + 5x + 1$ we get: 1

(a) $8x + 8$ **(b)** $8x + 7$ **(c)** $15x + 8$ **(d)** $16x$

6. The area of a parallelogram with base 11 cm and height 6 cm is: 1

(a) 34 cm² **(b)** 17 cm² **(c)** 33 cm² **(d)** 66 cm²

7. The number of vertices of a pentagonal pyramid is: 1

(a) 5 **(b)** 6 **(c)** 10 **(d)** 7

8. The range of 23, 26, 52, 35, 8, 28 is: 1

(a) 28 **(b)** 44 **(c)** 52 **(d)** 60

9. The two-digit number with tens digit x and ones digit 3 is written as: 1

(a) $x + 3$ **(b)** $3x$ **(c)** $30 + x$ **(d)** $10x + 3$

10. The area of a trapezium with parallel sides 10 cm and 4 cm and height 6 cm is: 1

(a) 120 cm² **(b)** 48 cm² **(c)** 84 cm² **(d)** 42 cm²

11. The number of edges of a pentagonal prism is: 1

(a) 10 **(b)** 12 **(c)** 7 **(d)** 15

12. The mean of 8 numbers is 12. Their sum is: 1

(a) 96 **(b)** 104 **(c)** 20 **(d)** 13

13. In a calendar, the four dates 6, 7, 13 and 14 form a 2×2 block. Their sum is: 1

(a) 42 **(b)** 38 **(c)** 40 **(d)** 32

14. The area of a rhombus with diagonals 12 cm and 9 cm is: 1

(a) 42 cm² **(b)** 21 cm² **(c)** 54 cm² **(d)** 108 cm²

15. A polyhedron has 5 faces and 5 vertices. By Euler's formula, its number of edges is: 1

(a) 8 **(b)** 12 **(c)** 9 **(d)** 10

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Simplify $2x + 3y + 5x + 4y$. 2

17. Find the area of a rhombus whose diagonals are 22 cm and 10 cm. 2

18. Write the number of faces, vertices and edges of a pentagonal prism. 2

19. Find the mean of 12, 48, 45, 41, 44. 2

20. Write an algebraic expression for: a number y multiplied by 4 and then reduced by 9. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Evaluate $5p + 2q - 8$ for $p = 4$, $q = 7$. 3

22. Find the area of a parallelogram with base 16 cm and height 7 cm. What is the area of the triangle formed by half of it (cut along a diagonal)? 3

23. In the Sierpinski triangle, how many shaded triangles are there at stage 3? How many more are there at stage 4? 3

24. Arrange the data 55, 58, 21, 65, 28, 54, 45, 47, 29 in ascending order and find its median. 3

OR

Simplify $12x + 3 - 7x + 6 - 4$. 3

25. Find the area of a trapezium whose parallel sides are 16 cm and 10 cm and the distance between them is 6 cm. 3

OR

A cuboid measures 8 cm by 4 cm by 3 cm. Find the total length of its edges and state its number of faces and vertices. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A Koch curve begins with a straight line of length 18 cm (stage 0). At each stage every segment is replaced by 4 segments each one-third as long. Find the number of segments, the length of each segment and the total length at stage 2. 5

OR

For the data 5, 7, 22, 14, 7, 26, find the mean, median, mode and range. 5

27. Think of a number n. Multiply it by 4, add 7 and then subtract 7. Show the result algebraically. What is the result when $n = 6$? 5

OR

A square plot has a perimeter of 32 m. Find its side and area. How many square tiles of side 1 m are needed to cover it? 5

END OF QUESTION PAPER — **Total: 50 Marks**